26900413 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEPHY304 - PHYSICS OF ADVANCED MATERIALS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Which are the different forms in which abrasives are used? [U] / [CO1]
2. Name the three typical types of stress–strain behaviour exhibited by polymeric materials. [U] / [CO1]
3. Give two examples of plastics and mention one important application of each. [U] / [CO1]
4. Define electron mobility. State its SI unit. [K] / [CO2]
5. How crazes propagate relative to tensile stress? [U] / [CO1]
6. Give two examples of structural clay products. [K] / [CO1]
7. What is meant by spontaneous polarization in a dielectric material? [U] / [CO2]
8. State any two important properties of Ultra-High-Molecular-Weight Polyethylene (UHMWPE). [U] / [CO1]

9. What is meant by the freeze-out region in an extrinsic semiconductor? [K] / [CO2]

10. Write a note about ferromagnetic materials? [U] / [CO3]

11. State the fundamental cause of thermal expansion in solids at the atomic/microscopic level. [U] / [CO4]

12. Why are highly amorphous polymers transparent, whereas highly crystalline polymers of the same composition are translucent or opaque? [U] / [CO5]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

13. Explain the two phases present in composites with example. [K] / [CO1]

14. Explain fracture toughness of ceramic materials. Write its expression and explain the terms involved. [U] / [CO1, CO2, CO3, CO4, CO5]

15. Explain the mechanism of electrical conduction in conjugated polymers. [U] / [CO2]

16. Explain why electronic thermal conduction predominates in high-purity metals. [U] / [CO4]

17. Discuss the wide range of electrical conductivities exhibited by solid materials, and explain how this range forms the basis for classifying materials electrically. [U] / [CO2]

18. Plot the saturation magnetization as a function of temperature for iron and Fe_3O_4 [K] / [CO3]

19. Explain the increase of capacitance using the simplified model of polarisation within a dielectric material using a schematic diagram. [A] / [CO2]

20. Explain any two optical properties of metals? [K] / [CO5]

[6x5 = 30]